

SIMON SCHLEICH

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RESEARCH INTEREST	Exoplanet atmospheric retrieval, data reduction of exoplanet observations, star-planet interactions and co-evolution, habitability.	
EDUCATION	PhD (Dr. rer. nat., Astronomy) University of Vienna, <i>Vienna, Austria</i>	2022 – present
	MSc (Master of Science, Astronomy) University of Vienna, <i>Vienna, Austria</i> Graduated with Distinction	2019 – 2022
	BSc (Bachelor of Science, Astronomy) University of Vienna, <i>Vienna, Austria</i>	2016 – 2019
RESEARCH EXPERIENCE	PhD project , University of Vienna THESIS TITLE: <i>Challenging atmospheric retrievals: Towards robust parameter inference from exoplanet transmission spectroscopy with next-generation telescopes</i> Characterising exoplanet atmospheres from JWST observations. End-to-end data reduction and application of atmospheric retrieval techniques, with a focus on NIR transmission spectroscopy.	10/2022 – ongoing
	MSc project , University of Vienna THESIS TITLE: <i>Stellar wind simulations in the age of PSP and Solar Orbiter</i> Investigated the properties of the solar wind using both numerical simulations and observation. The project included (1) implementation of a WTD heating mechanism into the 3D MHD code NIRVANA (<i>code development</i>), and (2) data analysis of PSP and SoLO measurements. Results have been published in Schleich et al. (2023) .	10/2020 – 01/2022
SELECT PUBLICATIONS	Schleich, S. , Boro Saikia, S., Changeat, Q., Güdel, M., Voigt, A. and Waldmann, I. (2025). <i>Knobs and dials of retrieving JWST transmission spectra. II. Impacts of pipeline-level differences on retrieval posteriors</i> . A&A, 704 , A223	
	Schleich, S. , Boro Saikia, S., Changeat, Q., Güdel, M., Voigt, A. and Waldmann, I. (2024). <i>Knobs and dials of retrieving JWST transmission spectra: I. The importance of p-T profile complexity</i> . A&A, 690 , A336 .	
	Schleich, S. , Boro Saikia, S., Ziegler, U., Güdel, M. and Bartel, M. (2023). <i>NIRwave: A wave-turbulence-driven solar wind model constrained by PSP observations</i> . A&A, 672 , A64 .	
SCIENTIFIC WORK	Scientific contribution: Poster Impacts of pipeline-level differences on atmospheric retrievals. Ariel Open Conference , Harwell, UK	2026
	Scientific contribution: Poster Sensitivity in atmospheric retrievals of JWST hot Jupiter transmission spectra. <i>European Astronomical Society (EAS) meeting</i> , Cork, Ireland	2025
	Scientific contribution: Talk Retrieving pressure-temperature profiles from exoplanet transmission spectra. <i>Ariel Consortium Meeting</i> , Lisbon, Portugal	2024

	Scientific contribution: Poster 2024 Influences of data processing techniques on the interpretation of atmospheric spectra from JWST. Exoplanets 5 , Leiden, The Netherlands.
	Scientific contribution: Talk 2024 Influences of data processing techniques on the interpretation of atmospheric spectra from JWST. Austrian Early Career Conference , Salzburg, Austria
	Scientific contribution: Poster 2023 NIRwave: A wave-turbulence-driven solar wind model constrained by Parker Solar Probe observations. Planet ESLAB 2023 , Leiden, The Netherlands. DOI: 10.5281/zenodo.7761620
OTHER WORK	LOC member: Science with the Hubble and James Webb Space Telescopes VIII 2026 ESA-sponsored conference , in collaboration with STScI, highlighting the scientific impact of the Hubble and James Webb Space Telescopes. Held in Vienna.
	SOC member: Wienerwald Astronomy Symposium 2024 Joint symposium between the Department of Astrophysics at the University of Vienna and the Astronomy Research Groups at the Institute for Science and Technology Austria (ISTA).
	Co-organisier: Big Picture Talks and Events 2023 [Big Picture Talk] Graveyard of Space Technology with approximately 100 participants (online and in-person)
OUTREACH	Speaker: Nights at the observatory (Nachts auf der Sternwarte) 2025 The sun, moon, and stars: Astronomy as a science and inspiration for music (Sonne, Mond und Stenre: Astronomie als Wissenschaft und Inspiration in der Musik , held in German). A recording is available here .
	Speaker: Nights at the observatory (Nachts auf der Sternwarte) 2025 New horizons on extrasolar planets (Neue Horizonte auf extrasolaren Planeten , held in German). A recording is available here .
	Speaker: Long Night of Research (Lange Nacht der Forschung) 2024 Observing exoplanets with space telescopes (Beobachtungen von Exoplaneten mit Weltraumteleskopen , held in German).
GRANTS	Grant for Early-stage Researcher - University of Vienna 2025 / 2024 / 2023 Granted for a poster contribution to EAS (2025, poster contribution), Exoplanets 5 (2024, poster contribution), PLANET ESLAB (2023, poster contributon)
	"International Communications" - Austrian Research Federation (ÖFG) 2023 Granted for a short-term research visit to the ESA office at STScI, Baltimore
	"Short-term research stay abroad (KWA)" - University of Vienna 2023 Granted for a short-term research visit to the ESA office at STScI, Baltimore
	Performance scholarship (Leistungsstipendium) - University of Vienna 2021 / 2020 Granted for excellence in academic achievement
SKILLS	Programming Python (experienced) C (familiar) HPC (Slurm Workload Manager)
	Software and Tools

[Eureka!](#) (JWST data reduction) | [TauREx3](#) (atmospheric retrieval) | $\LaTeX$ | ParaView

Languages

German (*Native*) | English (*Fluent*)

**List of
publications**

Robeling, N.-M., Boro Saikia, S., Van Looveren, G., Stankovic, I., **Schleich, S.**, Johnstone, C., Güdel, M., and Kislyakova, K. (submitted). *Thin on Dimensions, Thick on Physics: Self-consistent 1D Modelling of Jupiter's Upper Atmosphere*. submitted to A&A.

Boro Saikia, S., Tennyson, J., Ji, S., Robeling, N.-M., Stankovic, I., Van Looveren, G., **Schleich, S.**, Johnstone, C., Kislyakova, K., and Güdel, M. (submitted). *The physical and chemical structure of Titan's upper atmosphere. Energy balance, photochemistry, and Jeans escape*. submitted to A&A.

Schleich, S., Boro Saikia, S., Changeat, Q., Güdel, M., Voigt, A. and Waldmann, I. (2025). *Knobs and dials of retrieving JWST transmission spectra. II. Impacts of pipeline-level differences on retrieval posteriors*. A&A, [704](#), [A223](#)

Van Looveren, G., Boro Saikia, S., Herbort, O., **Schleich, S.**, Güdel, M., Johnstone, C. and Kislyakova, K. (2025). *Habitable Zone and Atmosphere Retention Distance (HaZARD): Stellar-evolution-dependent loss models of secondary atmospheres*. A&A, [694](#), [A310](#).

Schleich, S., Boro Saikia, S., Changeat, Q., Güdel, M., Voigt, A. and Waldmann, I. (2024). *Knobs and dials of retrieving JWST transmission spectra: I. The importance of p - T profile complexity*. A&A, [690](#), [A336](#).

Schleich, S., Boro Saikia, S., Ziegler, U., Güdel, M. and Bartel, M. (2023). *NIRwave: A wave-turbulence-driven solar wind model constrained by PSP observations*. A&A, [672](#), [A64](#).